

Simulation Program to Study Zener Diode Characteristics

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Testing electronic circuits directly on a breadboard or PCB may damage the electronic components, especially if the inputs to the circuit or connections are wrong. So students can first experiment with computer-based training (CBT) before doing the actual circuit implementation on the breadboard.

Here is a CBT program to study the working of a zener diode. The program is written in Visual Basic and is easy to understand and implement.

Software program

Three forms are used in this program: form1.frm for setup, frm_charzp.frm for characteristics and help.frm for 'help' menu. Some important program functions are explained below.

The Form_Load() function loads the width, height and positions of form1 and the various zener diode part numbers in the combo1 box of form1.

The VScroll1_Change() function is used to vary the input supply from 0 to 20 in steps of 0.1. If the user tries to change the input without entering the value of the resistance or without selecting the zener diode, a warning message is displayed.

The specifications of each zener diode are stored in 'zener' array in the code. The specifications include reverse leakage

current of the zener diode (I_o), reverse breakdown voltage of the zener diode (V_z) and maximum zener current (I_{Zmax}).

The current and voltage across the

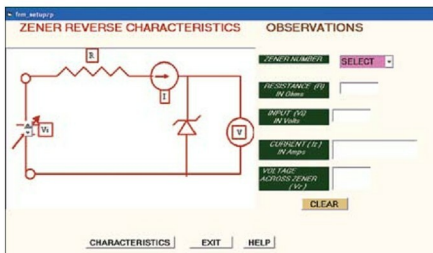


Fig. 1: Screenshot of program output

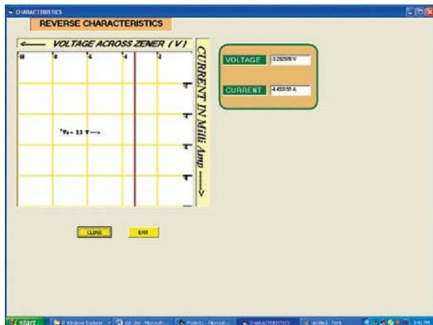


Fig. 2: Zener diode characteristics

Zener Diode Ratings

Zener diode	Voltage rating
1N4728A	3.3V
1N4729A	3.6V
1N4730A	3.9V
1N4731A	4.3V
1N4732A	4.7V
1N4733A	5.1V